SIMATS ENGINEERING

CO1 ASSESSMENT TOOL 1

COURSE NAME: Theory of Computation COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.

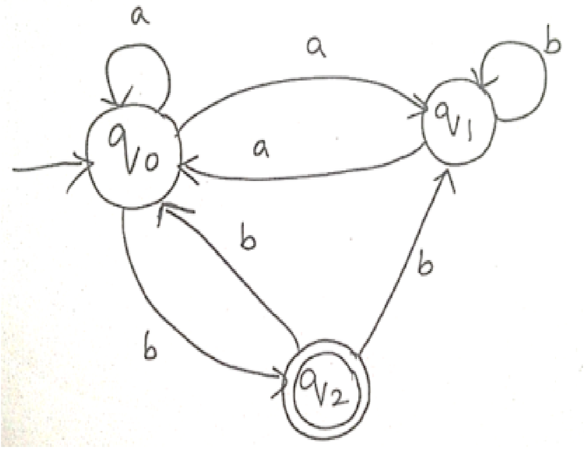
Analytical Problem

Code of Conduct:

I **GIRIJA B 192311044** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Girija B

1.Consider the NFA given below which accepts some language L over the alphabet Σ={a,b}. Construct a DFA that represents the same language L



Given NFA:

- **States:** $Q = \{q_0, q_1, q_2\}$
- **Alphabet:** $\Sigma = \{a, b\}$
- **Start State:** q_0
- **Final State:** q_2

Step 1: NFA Transition Table

State	**a**	**b**
$\rightarrow q_0$	$\{q_0,q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$*q_2$	$\{\ \}$	$\{q_0,q_1\}$

Step 2: Subset Construction

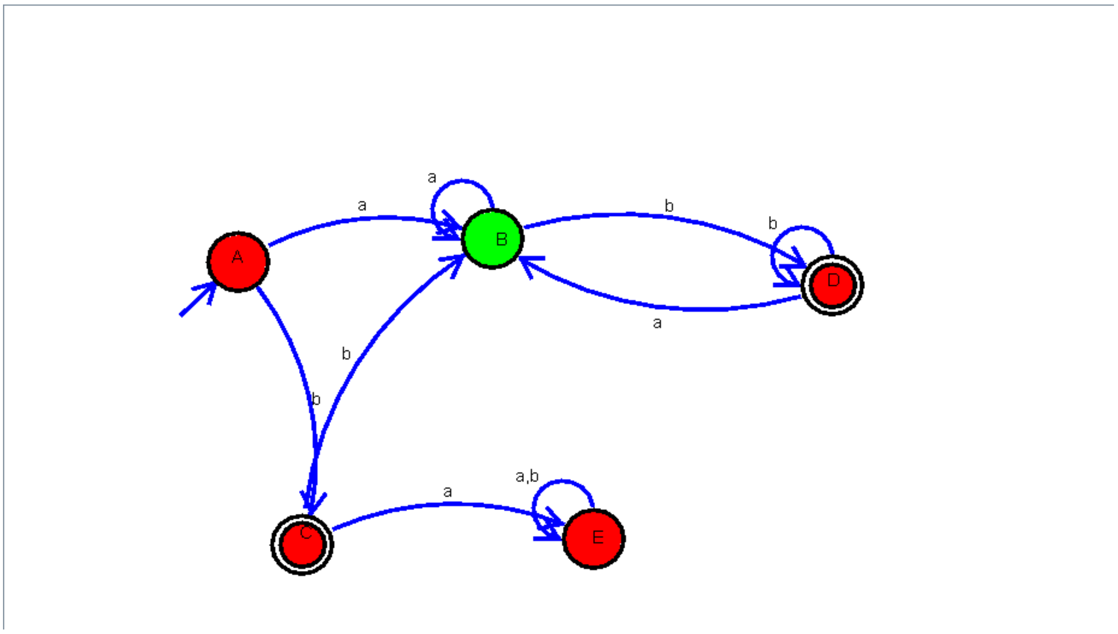
Let

- **A = $\{q_0\}$** (Start State)
- **B = $\{q_0,q_1\}$**
- **C = $\{q_2\}$** *(Final State)*
- **D = $\{q_0,q_1,q_2\}$** *(Final State)*
- **E = ∅** (Dead State)

Step 3: DFA Transition Table

DFA State	NFA States	a	b	Final
→A	$\{q_0\}$	B	C	No
B	$\{q_0,q_1\}$	B	D	No
*C	$\{q_2\}$	E	B	Yes
*D	$\{q_0,q_1,q_2\}$	B	D	Yes
E	∅	E	E	No

Step 4: DFA State Diagram



Final DFA

- **States:** {A, B, C, D, E}
- **Alphabet:** {a, b}
- **Start State:** A = $\{q_0\}$
- **Final States:** C = $\{q_2\}$, D = $\{q_0,q_1,q_2\}$

Conclusion

The DFA is obtained using the **subset construction method**, where each DFA state represents a set of NFA states. The constructed DFA accepts exactly the same language as the given NFA while ensuring deterministic transitions for every input symbol.

2.Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA.

Step 1: Language Description

$$L = \{ w \in \{a, b\}^* \mid |w| \geq 2 \text{ and last two symbols are different} \}$$

Accepted endings:

- **ab**
- **ba**

Rejected endings:

- **aa**
- **bb**

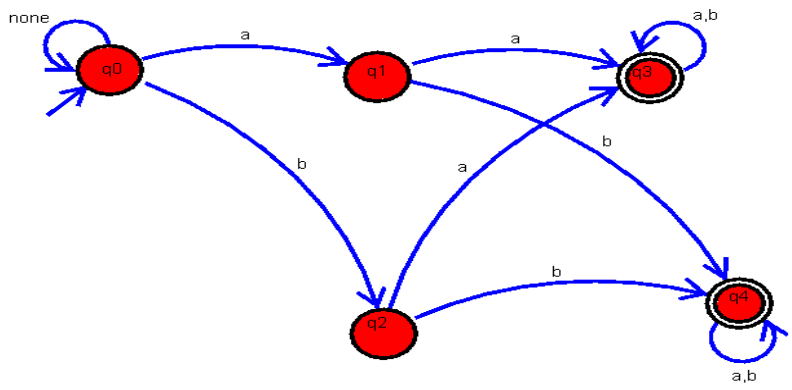
Step 2: NFA

States

- q_0 – Start state
- q_1 – Last symbol is **a**
- q_2 – Last symbol is **b**
- q_3 – Last two symbols are **ba** (Final)
- q_4 – Last two symbols are **ab** (Final)

Start State: $\mathbf{q_0}$

Final States: $\mathbf{\{q_3, q_4\}}$



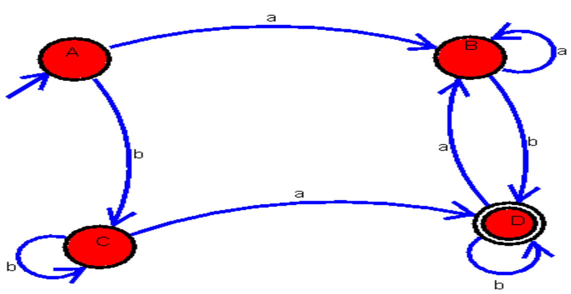
NFA Transition Table

State	a	b
$\rightarrow q_0$	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{\ \}$	$\{q_4\}$
q_2	$\{q_3\}$	$\{\ \}$
$*q_3$	$\{q_3\}$	$\{q_3\}$
$*q_4$	$\{q_4\}$	$\{q_4\}$

Step 3: Equivalent DFA

Let

- **A = $\{q_0\}$**
- **B = $\{q_0,q_1\}$**
- **C = $\{q_0,q_2\}$**
- **D = $\{q_0,q_2,q_4\}$** *(Accepting)*
- **E = $\{q_0,q_1,q_3\}$** *(Accepting)*



DFA Transition Table

DFA State	a	b	Final
→A	B	C	No
B	B	D	No
C	E	C	No
*D	E	C	Yes
*E	B	D	Yes

Step 4: DFA States

DFA State	Represents
A	$\{q_0\}$
B	$\{q_0,q_1\}$
C	$\{q_0,q_2\}$
D	$\{q_0,q_2,q_4\}$
E	$\{q_0,q_1,q_3\}$

Step 5: Final Answer

- **Start State:** A
- **Final States:** D, E

Conclusion

The NFA recognizes strings ending in **ab** or **ba**. Using the **subset construction method**, an equivalent DFA is obtained with deterministic transitions. Both automata accept the same language:

$$L = \{\, w \in \{a, b\}^* \mid |w| \geq 2 \text{ and the last two symbols are different} \,\}$$

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3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA

Step 1: Language Description

Alphabet:

$$\Sigma = \{0, 1\}$$

Language:

$$L = \{\, w \in \{0, 1\}^* \mid \text{the last two bits are 00 or 11} \,\}$$

Step 2: NFA Design

States

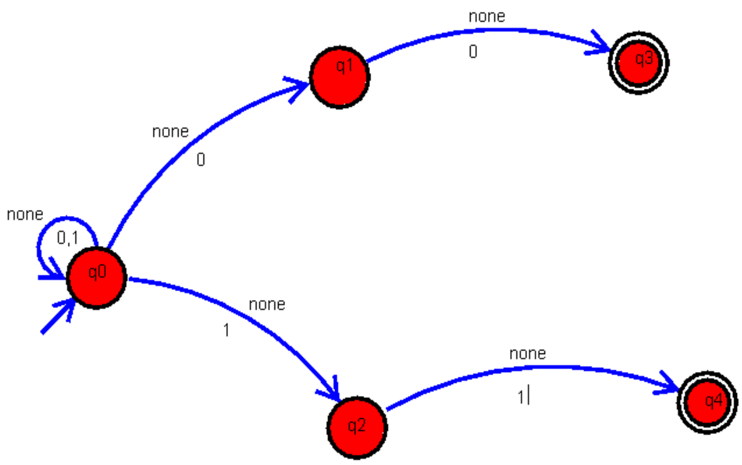
- q_0 – Start state
- q_1 – Last symbol guessed as 0

- q_2 – Last symbol guessed as 1
- q_3 – Accept state for ending 00
- q_4 – Accept state for ending 11

Start State: q_0

Final States:

$$F = \{q_3, q_4\}$$



NFA Transition Table

State	0	1
$\rightarrow q_0$	$\{q_0,q_1\}$	$\{q_0,q_2\}$
q_1	$\{q_3\}$	$\emptyset$
q_2	$\emptyset$	$\{q_4\}$
$*q_3$	$\emptyset$	$\emptyset$
$*q_4$	$\emptyset$	$\emptyset$

Step 3: Formal Definition

The NFA is defined as

$$M = (Q, \Sigma, \delta, q_0, F)$$

where

Set of States

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

Alphabet

$$\Sigma = \{0, 1\}$$

Transition Function

δ	0	1
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_3\}$	$\emptyset$
q_2	$\emptyset$	$\{q_4\}$
q_3	$\emptyset$	$\emptyset$
q_4	$\emptyset$	$\emptyset$

Start State : q_0

Final States

$$F = \{q_3, q_4\}$$

Step 4: Show that 01011 is Accepted

Input string: 01011

One accepting computation path is:

Input Read	Current State
Start	q_0
0	q_0
1	q_0
0	q_0
1	q_2 *(guess last two symbols start here)*
1	q_4 *(Accept)*

Since the computation ends in q_4, which is a final state,

$$\boxed{01011 \text{ is Accepted}}$$

because its last two digits are 11.

Conclusion

The designed NFA accepts all binary strings whose last two digits are the same (00 or 11). The formal definition $M = (Q, \Sigma, \delta, q_0, F)$, transition table, and accepting computation for 01011 verify that the NFA correctly recognizes the required language.

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4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet Σ={0,1}. Construct a DFA that represents the same language.

Step 1: Language Description

Alphabet:

$$\Sigma = \{0, 1\}$$

Language:

$$L = \{\, w \in \{0, 1\}^* \mid w \text{ contains 001 as a substring} \,\}$$

Step 2: Given NFA

States

- **q_0** – Start state
- **q_1** – First **0** detected
- **q_2** – Two consecutive **00** detected
- **q_3** – Substring **001** found (Final)

Start State:

$$q_0$$

Final State:

$$F = \{q_3\}$$

NFA Transition Table

State	**0**	**1**
→q_0	$\{q_0,q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	∅
q_2	∅	$\{q_3\}$
*q_3	$\{q_3\}$	$\{q_3\}$

Step 3: Subset Construction

Let

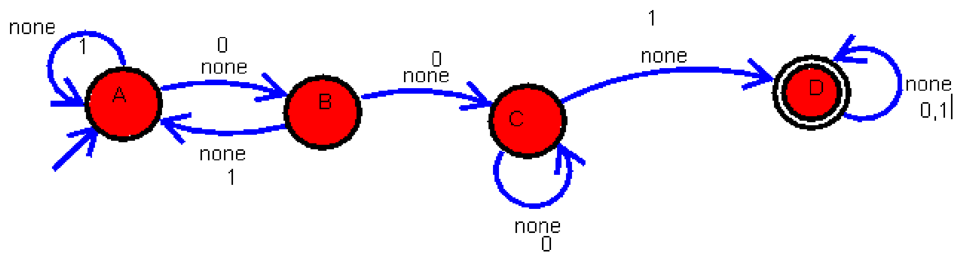
- **A = {q_0}**
- **B = {q_0,q_1}**
- **C = {q_0,q_1,q_2}**
- **D = {q_0,q_3}**

Since **D contains q_3**, it is an accepting state.

Step 4: DFA Transition Table

DFA State	NFA States	0	1	Final
→A	{q_0}	B	A	No
B	{q_0,q_1}	C	A	No
C	{q_0,q_1,q_2}	C	D	No
*D	{q_0,q_3}	D	D	Yes

Step 5: DFA Diagram



Step 6: DFA States Meaning

DFA State	Meaning
A	No part of **001** found
B	Last symbol is **0**
C	Last two symbols are **00**
D	Substring **001** has been found

Step 7: Final DFA

- **States:** {A, B, C, D}
- **Alphabet:** {0,1}
- **Start State:** A
- **Final State:** D

Conclusion

The DFA is constructed from the given NFA using the **subset construction method**. It accepts all binary strings that contain **001** as a substring. Once the substring **001** is detected, the automaton reaches the accepting state **D** and remains there for any further input.

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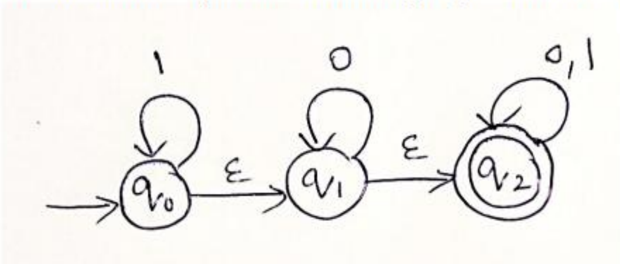
5. A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



Step 1: Given ε-NFA

States

- q_0 – Start state
- q_1
- q_2 – Final state

Transitions

- $q_0 \xrightarrow{1} q_0$
- $q_0 \xrightarrow{\varepsilon} q_1$
- $q_1 \xrightarrow{0} q_1$
- $q_1 \xrightarrow{\varepsilon} q_2$
- $q_2 \xrightarrow{0,1} q_2$

Step 2: ε-Closures

ε-closure(q_0)

From q_0:

- q_0
- $\varepsilon \rightarrow q_1$

- $\varepsilon \rightarrow q_2$

Therefore,

$$\boxed{\varepsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}}$$

ε-closure(q_1)

From q_1:

- q_1
- $\varepsilon \rightarrow q_2$

Therefore,

$$\boxed{\varepsilon\text{-closure}(q_1) = \{q_1, q_2\}}$$

ε-closure(q_2)

Only q_2 is reachable.

Therefore,

$$\boxed{\varepsilon\text{-closure}(q_2) = \{q_2\}}$$

Step 3: Construct Equivalent DFA

Let

- **A = $\{q_0, q_1, q_2\}$** *(Start State)*
- **B = $\{q_1, q_2\}$**
- **C = $\{q_2\}$**

Since every set contains **q_2**, all DFA states are **accepting states**.

Step 4: DFA Transition Table

From A = $\{q_0, q_1, q_2\}$

On **0**

- $q_0 \rightarrow$ —
- $q_1 \rightarrow q_1$
- $q_2 \rightarrow q_2$

ε-closure = **$\{q_1, q_2\}$ = B**

On **1**

- $q_0 \rightarrow q_0$

- $q_2 \rightarrow q_2$

ε-closure = **{q_0,q_1,q_2} = A**

From B = {q_1,q_2}

On **0**

- $q_1 \rightarrow q_1$
- $q_2 \rightarrow q_2$ = **B**

On **1**

- $q_2 \rightarrow q_2$ = **C**

From C = {q_2}

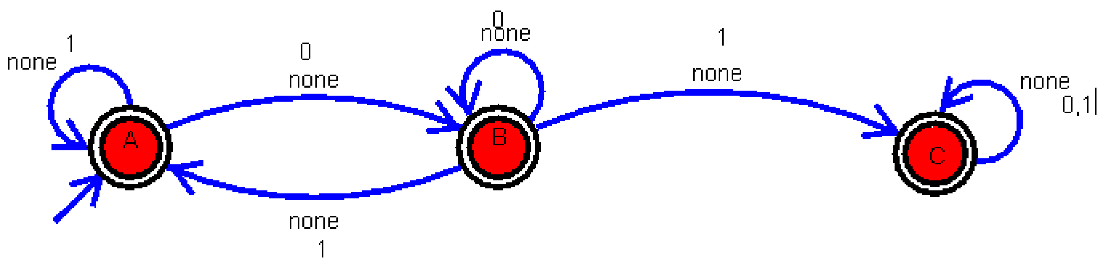
On **0**= **C**

On **1**= **C**

Final DFA Transition Table

DFA State	NFA States	0	1	Final
→*A	{q_0,q_1,q_2}	B	A	Yes
*B	{q_1,q_2}	B	C	Yes
*C	{q_2}	C	C	Yes

Step 5: DFA



Step 6: Initial and Final States

Initial State

$$\boxed{A = \{q_0, q_1, q_2\}}$$

Final States

$$\boxed{A,\ B,\ C}$$

(All contain the NFA final state q_2.)

Conclusion

The ε-closures are:

- ε-closure(q_0) = {q_0, q_1, q_2}
- ε-closure(q_1) = {q_1, q_2}
- ε-closure(q_2) = {q_2}

Using these ε-closures, the equivalent DFA has **three states (A, B, C)**. Since every DFA state contains the NFA final state **q_2**, **all DFA states are accepting states**. This DFA accepts the same language as the given ε-NFA.